01- Complex

Difficulty: Easy Expect Time: 30 min **Homework**

Define a class named Complex to represent complex numbers. A complex number is a number of the form $a + b * i$, where a and b are of type double, and i represents $\sqrt{-1}$.

- **Complex**

  The Complex class represents a complex number using two values of type double: one for the real part and one for the imaginary part.

  - **`realValue`**
  - **`imaginaryValue`**

  The Complex class includes three constructors:

  - **`Complex()`**
    Constructs a complex number with both **`realValue`** and **`imaginaryValue`** set to 0.
  - **`Complex(double r)`**
    Constructs a complex number with both **`realValue`** set to **`r`** and **`imaginaryValue`** set to 0.
  - **`Complex(double r, double i)`**
    Constructs a complex number with both **`realValue`** set to **`r`** and **`imaginaryValue`** set to **`i`**.

  The Complex class includes the following member functions:

  - **`double real()`**
    Returns the **`realValue`**.
  - **`double imag()`**
    Returns the **`imaginaryValue`**.
  - **`double norm()`**

    Returns the value of $\sqrt{(realValue^2 + imaginaryValue^2)}$.

  - **`friend double real(Complex c)`**
    Returns the **`realValue`** of the complex number **`c`**.
  - **`friend double imag(Complex c)`**
    Returns the **`imaginaryValue`** of the complex number **`c`**.
  - **`friend double norm(Complex c)`**

    Returns the value of $\sqrt{(realValue^2 + imaginaryValue^2)}$ of the complex number **`c`**.

The following operators are required to be overloaded:

- **`operator!=`**
  Compares two complex numbers to determine if both the real and imaginary parts are NOT equal.

- **`operator==`**
  Compares two complex numbers to determine if both the real and imaginary parts are equal.

- **`operator+`**
  Adds another complex number or a double value to the current complex number.

$$(a + bi) + (c + di) = (a + c) + (b + d)i$$

- **`operator-`**
  Subtracts another complex number or a double value from the current complex number.

$$(a + bi) - (c + di) = (a - c) + (b - d)i$$

- **`operator*`**
  Multiplies the current complex number by another complex number or a double value.

$$\begin{aligned}(a + bi)(c + di) &= ac + adi + bci + bdi^2 \\ &= ac + adi + bci - bd \\ &= (ac - bd) + (ad + bc)i\end{aligned}$$

- **`operator/`**
  Divides the current complex number by another complex number or a double value.

$$\begin{aligned}\frac{a + bi}{c + di} &= \frac{a + bi}{c + di} \times \frac{c - di}{c - di} \\ &= \frac{(a + bi)(c + di)}{c^2 + d^2} \\ &= \frac{(ac + bd)(bc - ad)i}{c^2 + d^2} \\ &= \frac{ac + bd}{c^2 + d^2} + \frac{bc - ad}{c^2 + d^2} i\end{aligned}$$

- **`operator>>`**
  Extracts the value of a complex number from the input stream in the format:

$$x = realValue + imaginaryValue * i$$

- **`operator<<`**
  Outputs the complex number to the output stream in the format:

$$realValue + imaginaryValue * i$$

Input

1. Complete the required class implementation. The following files already contain some implementations
   a) **`Complex.h`**
   You can submit a header-only implementation, or split the implementation into **`*.cpp`** files.

2. The Online Judge system will replace the following files:
   a) **`entrypoint.cpp`**
   b) **`case.h`**
   c) **`test.h`**

3. The test cases for student available in **case.h**, but not all test data is included in that file. The Online Judge system will also include hidden test inputs.

4. All test cases guarantee that **no division by zero will occur.** For all division operations, the divisor (whether a double or a Complex number) will be non-zero.

Output

Online Judge will run the test:

1. **DON' T** output any information to the terminal.
2. When any test failed, Online Judge will try to list error messages.